FEDERAL SPECIFICATION

CANNULA SET, INTRAVENOUS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 *This specification covers* a set of four sizes of woven venous cannulae, suitable for intravenous infusion purposes.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP–B–566—Boxes, Folding, Paperboard.
PPP–B–585—Boxes, Wood, Wirebound.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Box, Fiberboard.
PPP–B–676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 66—Steel: Chemical Composition and Hardenability.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–P–116—Preservation, Methods of.
MIL–L–10547—Liners, Case, Waterproof.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

(Copies of Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

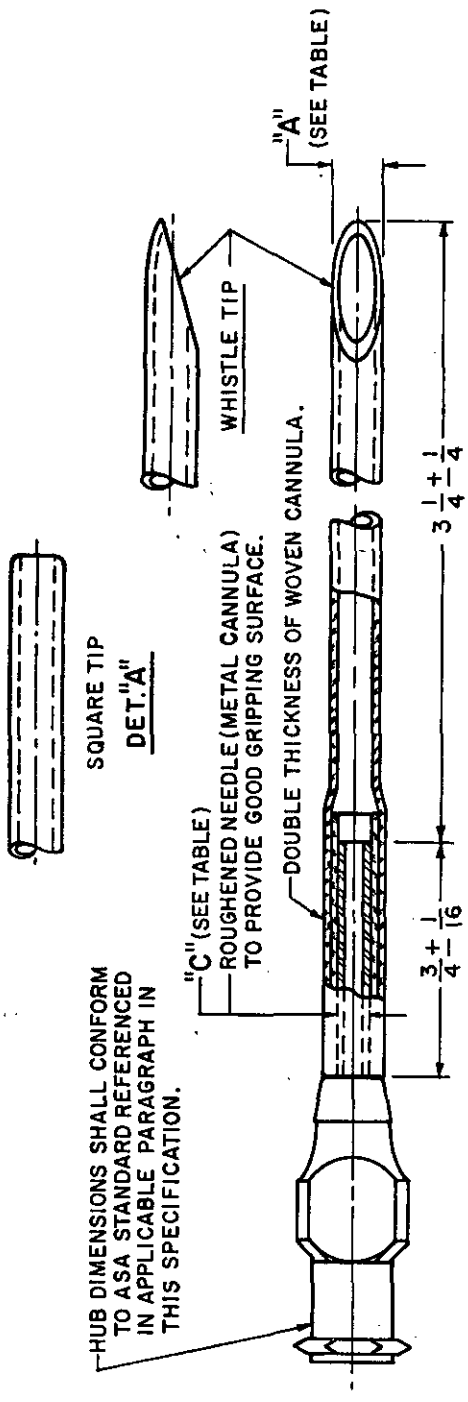
2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Materials Specification:

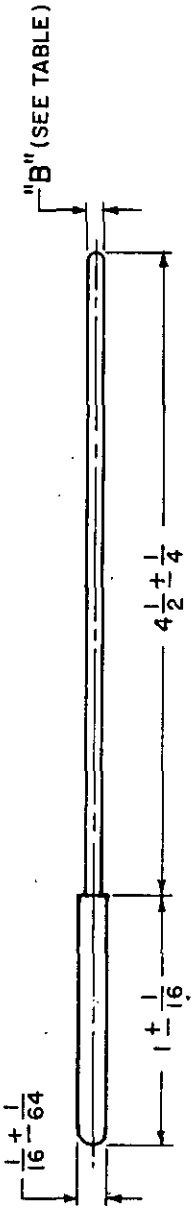
E1—ASTM Thermometers.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.



CANNULA



OBTURATOR

GEN. NOTES

THE 5, 6 & 7 FRENCH CANNULAE SHALL HAVE WHISTLE TIPS (AS SHOWN). THE 3.5 FRENCH CANNULA SHALL HAVE A SQUARE TIP. (SEE DET. "A".)

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.

A VENOUS CANNULA	B ±.001 (INCH) OBTURATOR SHAFT	C NEEDLE METAL CANNULA
3.5 FRENCH	.016 DIA.	20 GA.
5 FRENCH	.020 DIA.	18 GA.
6 FRENCH	.030 DIA.	17 GA.
7 FRENCH	.037 DIA.	16 GA.

FIG. 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

American Standards Association, Inc., Standard:

Z70.1–1955 (UDC615.47)—Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American Standards Association, Inc., 70 East 45th Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Woven tube.*

3.1.1.1 *Base fabric.* The base fabric of the woven tube shall be woven from high-tenacity nylon yarn of a size appropriate for the diameter of the cannula shaft. The yarn shall have relatively high absorbent properties so that the maximum adhesion is secured between the base and coating. The nylon yarn shall have a melting point of not less than 135° Centigrade (C.). Melting point shall be determined as specified in 4.3.1.1.

3.1.1.2 *Coating.* The coating shall be a varnish, or suitable synthetic or plastic compound.

3.1.2 *Needle.*

3.1.2.1 *Metal Cannula.* Metal cannula shall be seamless drawn, or welded and drawn tubing fabricated of identification No. AISI 304 stainless steel in accordance with the requirements of Federal Standard No. 66. The metal cannula shall be capable of passing the corrosion-resistance test specified in 4.3.2.1.1.

3.1.2.2 *Hub.* The needle hub shall be brass, plated with chromium or nickel, and polished, and shall be capable of passing the corrosion-resistance test specified in 4.3.2.1.2.

3.1.3 *Obturator.* The obturator shall be solid nylon.

3.2 Construction. The venous cannula shall consist of the following four sizes of cannulae: 3.5, 5, 6, and 7 French (Fr.). Each venous cannula shall consist of a woven tube attached to the metal cannula of a hypodermic needle and shall be furnished with an obturator. The cannulae, with obturators, shall be constructed as shown on figure 1, and as hereinafter specified.

3.2.1 Sizes. The size of the obturator and the size of the metal cannula for each size venous cannula shall be as listed in table I.

TABLE I.

Venous cannula (French)	Obturator shaft diameter (inch, tolerance ±0.001in.)	Needle cannula size (Gage)
3.5	0.016	20
5	.020	18
6	.030	17
7	.037	16

The French size of the woven tube shall be determined in accordance with the procedure specified in 4.3.1.2.

3.2.2 *Woven tube.* The woven tube shall be constructed as shown on figure 1. The tube shall be composed of a tightly and smoothly woven base, free from loose ends or broken fibers, and shall be completely, uniformly, and symmetrically covered with a flexible coating. The finished woven tube shall have proper flexibility, pliability, and body required to withstand pressure of insertion into veins, and shall have the softness and pliability necessary to prevent irritation to body tissues. The woven tube shall be 4 inches in overall length and of French size as indicated above (a set consisting of 4 venous cannulae, one venous cannula of each size). The portion of the woven tube from slightly past the distal end of the needle metal cannula (on the finished, assembled venous cannula) to the distal end of the woven tube shall be of appropriate French size for the venous cannula component of the set. The portion of the woven tube attached to the needle metal cannula, plus $\frac{1}{16}$ inch, on the finished assembled venous cannula shall be double-layer and of such diameter as necessary to accommodate appropriate gage needle to which

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

the woven tube is attached (as listed in table I for appropriate French size cannula).

3.2.2.1 *Base fabric.* The base shall be a single web (except proximal end of the woven tube, as described in 3.2.2, which shall be double web) tightly and smoothly woven in braided form on a 26 to 40 carrier machine with yarn of appropriate size and type for the size of the tube. The thickness of the wall shall be relatively uniform throughout the particular portion of the shaft (i.e., uniform double-layer wall throughout the proximal end of the woven tube and uniform single-layer wall throughout the distal end of the woven tube). Each yarn woven into the tube shall be of one continuous length. The woven tube shall have the required body to be resistant to buckling or breaking when tested as specified in 4.3.1.3, 4.3.1.4, 4.3.1.5, 4.3.1.6, and 4.3.1.7.

3.2.2.2 *Coating.* The coating shall be thoroughly and uniformly impregnated in and around the woven base in a manner to coat the braided fibers completely. The coating thickness shall be relatively uniform throughout the length of the woven tube. The coating shall be formulated to secure maximum adhesion to the base fabric and shall have the required toughness and elasticity to permit repeated flexing and to prevent peeling, cracking, or breaking when tested as specified in 4.3.1.4, 4.3.1.5, 4.3.1.6, and 4.3.1.7. The coating shall be resistant to body fluids, mild acids, alkalies, and deterioration due to aging during storage. The coated woven tube shall show no signs of brittleness, blistering, or other visible damage as a result of the boiling or autoclaving tests specified in 4.3.1.4 and 4.3.1.5. The finished tube shall not be tacky or stick together, and shall not show any other evidence of damage as a result of the flexibility tests specified in 4.3.1.7.

3.2.2.3 *Lumen.* The lumen of the woven tube shall be such that when tested as specified in 4.3.1.3 a mandrel of the size specified in table II for the appropriate size woven tube shall be capable of being passed completely through the woven tube.

TABLE II.

Venous cannula (French)	Mandrel size for testing lumen (inch)
3.5	0.018
5	.024
6	.033
7	.039

3.2.2.4 *Tip.* The extreme distal end of the 5 Fr., 6 Fr., and 7 Fr. cannulae shall have a whistle tip as shown on figure 1. The 3.5 Fr. cannula shall have a square-cut tip. The tips shall be formed during the weaving of the cannula shafts, being smoothly and symmetrically woven into the structure of the woven tubes, and shall have smooth beveled edges.

3.2.3 *Needle.*

3.2.3.1 *Metal cannula.* The metal cannula of the needle shall be ¾ inch in length. The metal cannula shall be without a point; the distal end of the metal cannula being cut square and smoothly finished. The other dimensions of the metal cannula shall be as specified in table III. The metal cannula shall be securely swaged or soldered into the hub of the needle (so as to be capable of passing tests for security of swaging (or soldering) and fluid tightness as specified in 4.3.2.2 and 4.3.3) and shall extend well into the hub and be concentric with the hub cavity. The metal cannula shall have a rough finish on the outer surface to facilitate cementing woven tube to the metal cannula and an inner surface free from scale, deep pits, deep furrows, die marks, chips, polisher's dirt, or other foreign material. When tested as specified in 4.3.2.1.1, the metal cannula shall show no appreciable corrosion.

3.2.3.2 *Hub.* The hub of the needle shall be constructed as shown on figure 3 of American Standards Association (ASA) Standard Z70.1–1955. The hub shall be so shaped that it will not roll and will provide satisfactory finger grip for attaching needles to a syringe. The entire hub cavity wall down to the entrance of the lumen shall be well plated. The hub cavity shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

TABLE III.

Venous cannula (French)	Needle metal cannula size (Gage)	Needle metal cannula outside diameter		Needle metal cannula inside diameter (lumen), measured at distal end, minimum *Inch*
		Maximum *Inch*	Minimum *Inch*	
3.5	20	0.036	0.034	0.022
5	18	.0505	.0475	.0315
6	17	.0595	.0565	.0365
7	16	.0665	.0635	.0415

be free of chips, polisher's dirt, or other foreign material when tested as specified in 4.3.2.3. When tested as specified in 4.3.2.1.2, the hub shall show no appreciable corrosion.

3.2.3.3 *Diameter and depth of the hub cavity.* Diameter and depth of the hub cavity shall be inspected as specified in 4.3.2.4. When so tested, the hub cavities shall fit the plug gage within the "GO" and "NO GO" limits of the feeler gage as shown on figures 5 and 6 of the ASA Standard Z70.1–1955. Hubs which do not fit the plug gage within these limits shall be considered not acceptable. In addition, when the hub is fitted into the plug as specified in 4.3.2.4, there shall be no movement of the hub on the gage when the needle cannula is pressed lightly from side to side in all directions.

3.2.4 *Obturator.* The solid-nylon obturator shall be constructed as shown on figure 1. The proximal 1 inch of the obturator shall be $\frac{1}{16}$ inch in diameter and shall serve as a handle for the obturator.

3.2.5 *Assembly.* The proximal end of the tube shall be fitted onto the metal cannula of its appropriate hypodermic needle, as specified in table I above, and shall be securely cemented to the metal cannula by means of a sterilizable nontoxic adhesive.

3.2.5.1 *Fluid tightness.* When tested for fluid tightness as specified in 4.3.3, the swaged (or soldered) connection of the needle metal cannula with the needle hub, the Luer connection of the needle hub to the plug gage, and the assembly of the woven tube to the metal cannula on the finished assembled venous cannula, shall not show leakage exceeding the formation of a falling drop.

3.2.5.2 *Security of assembly.* The securing of the woven tube to the metal cannula shall be such as to pass the test specified in 4.3.4.

3.2.6 *Color code.* The color code of the outer surface of the finished woven tube shall be brown. A color-code system shall be used for the venous-cannula set to facilitate determination of the particular size venous cannula of each set. Approximately 1 inch of the proximal end of each woven tube shall have a color band around woven tube. The color of this band shall be as specified in table IV for the particular size venous cannula. The obturator handle shall be of the same color as the color of the woven tube with which it mates to form a unit.

TABLE IV.

Venous cannula (French)	Obturator shaft diameter (tolerance ±0.001 in.)	Color code
	Inch	
3.5	0.016	Black
5	.020	Gray
6	.030	Green
7	.037	Orange

3.3 Toxicity. The venous cannulae shall be free from toxic material or any ingredient that might have a deleterious effect upon body tissues.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.4 Finish. The surface of each woven tube shall be properly worked down and given a dry, smooth, highly polished but elastic finish, free from scratches, grooves, or other defects which would prevent the cannula from being readily inserted into veins. The surface shall be "slick" when wet and pass easily.

3.5 Identification marking. Each venous cannula shall be plainly, legibly, and permanently marked on the color band of the woven tube with the name or registered trademark of the manufacturer and the French size of the particular venous cannula.

3.6 Workmanship. Workmanship shall be first class throughout. Venous-cannula set shall be free from defects which detract from its appearance or may impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* When available, certificates of quality supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* The inspection lot shall be determined by the cognizant government inspector and shall be based on the unit of product being one cannula (end item), and shall be defined (other than for packaging, packing, and marking) in terms of the number of cannulae of the same size.

4.2.2 *Sampling for examination.* A random sample of cannulae shall be selected from each lot offered to the Government in accordance with Military Standard MIL–STD–105. Examination for dimensions shall be in accordance with Military Standard MIL–STD–105, and as indicated in table V.

4.2.3 *For tests.* Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105, and as indicated in table V.

4.2.4 *Raw materials.* Four ounces of each lot of each metal used in the construction of the cannulae shall be selected for chemical analysis.

4.2.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.3.1 *Woven tube.*

4.3.1.1 *Melting point.* A piece of nylon yarn, approximately ⅛ inch long, taken from the base, shall be placed in a capillary tube closed at the lower end. The piece of yarn shall be attached to a partial immersion thermometer and immersed to the proper depth in a bath at a temperature of 135°C. for 5 minutes. Thermometer shall conform to designation 2C (–5° to +300°C.), of American Society for Testing Materials Specification E1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

TABLE V.—*Sampling criteria*

Component	Characteristic	Req. par.	Test procedure	Insp. level	AQL (percent defective)
Base fabric	Melting point	3.1.1.1	4.3.1.1	*	*
Cannula	Outside diam.	3.2.1	4.3.1.2	II	1.0
Woven tube	Mandrel test	3.2.2.3	4.3.1.3	L2**	***
	Boiling test	3.2.2.2	4.3.1.4	L2**	***
	Autoclaving test	3.2.2.2	4.3.1.5	L2**	***
	Alcohol immer.	3.2.2.2	4.3.1.6	L2**	***
	Flexibility—wet	3.2.2.2	4.3.1.7.1	L2**	***
	Flexibility—dry	3.2.2.2	4.3.1.7.2	L2**	***
Needle	Corr. resist.—cannula	3.2.3.1	4.3.2.1.1	L2**	***
	Corr. resist.—hub	3.2.3.2	4.3.2.1.2	L2**	***
	Security of swaging	3.2.3.1	4.3.2.2	L5	1.5
	Freedom from foreign matter	3.2.3.2	4.3.2.3	L5	1.5
	Diam. and depth	3.2.3.3	4.3.2.4	II	1.0
	Fluid tightness	3.2.5.1	4.3.3	L7	1.5
Cannula	Security of assemb.	3.2.5.2	4.3.4	L5	1.5
	Dimensions	3.2	.....	II	4.0

* Each lot (or batch) of material shall be tested.
** A minimum of two units shall be tested regardless of lot size.
*** Acceptance number shall be zero for all sample sizes.

4.3.1.2 *Outside diameter.* The outside diameter of the distal end of the woven tube (from the distal end of the metal cannula on the finished, assembled venous cannula, to the distal end (tip) of the woven tube) shall be tested for French size against a standard French scale. The cannula shaft being checked shall pass through the specific scale opening of the size designated, but not through the next smaller size opening of the French scale.

4.3.1.3 *Mandrel test.* The woven tube shall be immersed for 3 hours in distilled water at room temperature. A mandrel of the size specified in table II for the appropriate size woven tube shall then be passed completely through the woven tube.

4.3.1.4 *Boiling test.* The woven tube shall be boiled for ten 20-minute intervals, with 24 hours between each boiling, to permit drying out at room temperature (68° to 72° Fahrenheit (F.)).

4.3.1.5 *Autoclaving test.* The woven tube shall be autoclaved for ten 10-minute intervals at 250°F., with sufficient time between to permit cooling to room temperature (68° to 72°F.).

4.3.1.6 *Alcohol immersion.* The woven tube shall be immersed in 70 percent alcohol for 5 minutes at room temperature (68° to 72°F.).

4.3.1.7 *Flexibility tests.*

4.3.1.7.1 *Wet test.* The woven tube shall be twisted three full-length turns around a woven tube of the same size, subjected to boiling for 20 minutes, allowed to return to room temperature, then untwisted.

4.3.1.7.2 *Dry test.* The woven tube shall be twisted three full-length turns around a woven tube of the same size, subjected to autoclaving for 15 minutes at 250°F., allowed to return to room temperature, then untwisted.

4.3.2 *Needle.*

4.3.2.1 *Corrosion resistance.*

4.3.2.1.1 *Metal cannula.* The metal cannula shall withstand immersion in 10 percent citric

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

acid at room temperature (68° to 72°F.) for 5 hours and after removal from the acid, boiling for 30 minutes in distilled water, followed by immersion in the distilled water at room temperature (68° to 72°F.) for 48 hours, without showing appreciable corrosion after drying by evaporation. Tests shall be made in a glass vessel.

4.3.2.1.2 *Hub.* The needle shall withstand boiling for 30 minutes and remaining immersed for 24 hours in distilled water without showing appreciable corrosion after drying by evaporation. Test shall be made in a glass vessel.

4.3.2.2 *Security of swaging (or soldering).* The swaging (or soldering) of the metal cannula into the hub shall be such that the metal cannula will stand 1 minute without pulling out of the hub the weight indicated in table VI. The swaging (or soldering) shall also withstand the test for fluid tightness as specified in 4.3.3.

TABLE VI.

Needle metal cannula size (Gage)	Pull to be applied for 1-minute (Pounds)
20	15
18	15
17	20
16	20

4.3.2.3 *Freedom from foreign matter in the hub cavity.* The needle hub cavity, under visual examination with a magnifier and good illumination, shall show complete freedom from metal chips and other foreign matter. The hub cavity shall also show freedom to the extent that a cotton swab tightly wrapped on the end of a small-diameter wooden applicator and vigorously turned down into the bottom of the cavity shall show no metal chips or other foreign matter on the cotton fibers when withdrawn and visually examined under a magnifier.

4.3.2.4 *Diameter and depth.* Hub cavity shall be tested on the male Luer taper gage in accordance with plug gage of figure 5 of the ASA Standard. This gage shall be as shown in the ASA Standard, except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test specified in 4.3.3 The hub shall be slipped onto this gage with very light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with figure 5 of the ASA Standard.

4.3.3 *Fluid tightness.* With a finished, assembled venous cannula (except for obturator) in a horizontal position, the hub shall be attached by means of a plug gage of figure 5 of the ASA Standard (except that gage shall have a suitable fluid path). The water shall be run through the needle so as to fill any hub-cavity space and the lumen of the needle and woven tube. The tip of the woven tube shall then be sealed, a pressure of 45 pounds per square inch applied for 10 seconds, and the assembled venous cannula examined for leakage.

4.3.4 *Security of assembly.* The finished, assembled venous cannula shall be pulled in a tensile strength machine. The woven tube shall break before the bond between this shaft and the metal cannula is broken. This security of assembly shall remain unchanged after venous cannulae have been autoclaved for 30 minutes at 250°F., and after they have been boiled for 1 hour.

4.4 Examination. The cannulae shall be examined to determine compliance with all requirements contained in this specification.

5. PREPARATION FOR DELIVERY

For civil agency procurement, Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment.

5.1 Packaging.

5.1.1 *Level A.*

5.1.1.1 *Unit package.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

5.1.1.1.1 *Cleaning.* Each unit shall be thoroughly cleaned in accordance with the applicable method as outlined in Military Specification MIL–P–116.

5.1.1.1.2 *Wrapping and cartoning.* Each unit shall be completely wrapped in neutral tissue or other suitable anticorrosive material and then packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Carton shall contain sufficient cushioning material to prevent shifting and breakage and shall be adequately secured to prevent accidental opening.

5.1.1.1.3 *Bag.* Each carton shall be further packaged in accordance with method IA–8 of Military Specification MIL–P–116.

5.1.1.2 *Intermediate package.* Twelve unit packages as specified in 5.1.1.1 shall be intermediately packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Carton shall be secured to prevent accidental opening.

5.1.2 *Level B.*

5.1.2.1 *Unit package.*

5.1.2.1.1 *Cleaning shall be as specified in 5.1.1.1.1.* Wrapping and cartoning as specified 5.1.1.1.2.

5.1.2.2 *Intermediate package.* Twelve unit packages as specified in 5.1.2.1 shall be intermediately packaged in a carton of appropriate size, or a complete wrapping of kraft paper of suitable weight. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3 *Level C.* Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection reqiured for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.* Twenty-four intermediate containers shall be packed in exterior containers designed for type 2 load and constructed in accordance with Federal Specification PPP–B–601, table II, when using type III, class 1 plywood; or PPP–B–621, table III. Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.* Twenty-four intermediate containers shall be packed in exterior containers conforming to Federal Specicatifion PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I, or PPP–B–636, type I, class I, fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.* Unless otherwise specified in the contract or order, unit, intermediate, and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Interior packages.* Each unit carton and barrier, when used, and each intermediate

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

package shall be marked in accordance with Military Standard MIL–STD–129. In addition, contract number and name of contractor shall appear.

5.3.2.2 *Exterior container.* Exterior container shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(1) Level B preservation and packaging is intended to provide economical but limited protection, and should be specified only when it is determined the cannula set will be held in covered storage approimatelxy 12 months from date of initial packing.

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Surgical instruments, not otherwise indexed by name.

Carload minimum weight 10,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Surgical instruments, not otherwise indexed.

Truckload minimum weight 10,000 pounds, subject to Rule 34, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

CUSTODIANS:

Army—M

Navy—Md

Air Force—AFCSG

Copies of this specification may be purchased for 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.